

MCR3M - M1C4

Agenda:

1. Action: Domain and Range of Quadratic Functions
2. Practice: GoFormative
3. Story 3 Introduction: My AI Learning Lab

Learning Goal:

I will identify domain and range of quadratic functions

Recall: Domain & Range

Domain:

Range:

Set Notation: A system to denote all items belonging to a set using curly brackets { }

Domain & Range of Linear Functions

Ex. 1: Identify the domain and range of

a) $f(x) = 2x - 3$

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b) $h(x) = 5$

Domain & Range of Quadratic Functions

Ex. 2: Identify the domain and range of $f(x) = -3(x - 1)^2 + 6$

What can we do to figure this out?

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Domain & Range of Quadratic Functions

Ex. 3: Clay shooting disks are launched from the ground into the air from a machine 12 m above the ground. The height of each disk, $h(t)$, in metres, is modelled by $h(t) = -5(t - 3)^2 + 57$, where t is the time in seconds since it was launched.

- a) What is the maximum height the disks reach?
- b) At what time do the disks hit the ground? Use table of values or graph (approx. one decimal place)
- c) Determine the domain and range of this model.

You Try It: Answer question 1 of today's M1C3 Formative

Identify the domain and range of

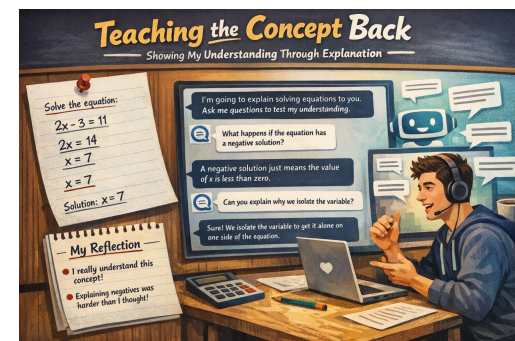
$$f(x) = 2(x + 4)^2 - 3$$

Story 3: My AI Learning Lab

Purpose: Show depth of understanding through explanation.

Required Components

- Choose **one (1) textbook question** you now understand well. Include an image of the question and your solution, including any work shown.
- Use the following prompt with an AI tool: "I'm going to explain [concept] to you. Ask me questions to test my understanding."
- **Include screenshots** of your prompt, explanation, and the AI's probing questions (at least two), including your responses
- **Reflection**
 - What did explaining this concept reveal about my understanding?
 - What was harder to explain than I expected?



Cheat Code: There isn't one! Just do your best and think critically as you interact

Tips:

- Attach an image of the textbook question with your prompt
- Use your microphone to record your teaching



Consolidation

GoFormative M1C3: Quadratic Domain & Range